

CameraLa: A Privacy-Preserving Web Framework for Unobtrusive Longitudinal Context-Aware Monitoring

Anonymous Author(s)
Anonymous Affiliation(s)

Abstract—Objective: As ubiquitous computing rapidly transitions to continuous, in-the-wild monitoring, it faces practical challenges regarding privacy and the interpretation of physiological signals across different environments. Traditional wearable models often rely on context-agnostic mappings, ignoring “Invariance Breaking”—where identical physiological patterns (e.g., elevated heart rate) represent divergent states depending on the environment (e.g., threat vs. engagement).

Methods: We propose CameraLa, a privacy-preserving, fully client-side web sensing framework designed for longitudinal affective monitoring. By processing high-resolution facial landmarks and remote photoplethysmography (rPPG) signals entirely within the browser via WebAssembly (Edge Computing), CameraLa enables ethically sound monitoring in private spaces without transmitting raw video. To evaluate its deployability and demonstrate the critical necessity of context-aware modelling, we conducted a proof-of-concept autoethnographic feasibility study ($N = 1,600$ trials) comparing two environments: a University Laboratory and a naturalistic Home setting.

Results: The system successfully operated in both environments, validating the edge-computing pipeline. Furthermore, the deployment case study demonstrated the system’s ability to capture “Invariance Breaking.” While a cognitive stressor induced physiological freezing and accelerated reaction times in the Lab, these metrics were statistically indistinguishable from baseline in the Home environment, highlighting the home’s role as a physiological buffer.

Conclusion: CameraLa demonstrates the feasibility of high-fidelity, privacy-preserving web sensing for ubiquitous computing. Our proof-of-concept findings highlight that future affective models must be dynamically parameterized by the user’s environmental context to properly interpret physiological signals in the wild.

Index Terms—Web-based Sensing, Single-Case Experimental Design, Invariance Breaking, Remote Photoplethysmography (rPPG), Affective Computing, Ecological Validity, Context Dependency, Polyvagal Theory, Challenge and Threat, Privacy-Preserving AI.

I. INTRODUCTION

THE vision of Affective Computing, as originally articulated by Picard within the MIT Media Lab in the late 1990s [1], was to endow machines with the capacity to recognize, interpret, simulate, and potentially influence human emotions. For over three decades, this vision has catalyzed extensive interdisciplinary research spanning computer science, psychology, neuroscience, and biomedical engineering. The ability to automatically detect user states—ranging

from frustration and stress to engagement and flow—promises to revolutionize human-computer interaction (HCI), enabling systems that adapt dynamically to the user’s cognitive and emotional needs.

Early work in this domain was predominantly conducted in highly controlled laboratory environments. Researchers utilized medical-grade sensors—such as electrocardiograms (ECG) to measure heart rate variability (HRV), electroencephalograms (EEG) to monitor cortical asymmetry, and galvanic skin response (GSR) sensors to track sympathetic arousal. These pioneering studies successfully established the somatic markers of basic emotions, enabling the development of classifiers that could distinguish high-arousal states (e.g., anger, fear) from low-arousal states (e.g., sadness, relaxation) with remarkable accuracy. Indeed, within the sanitized, noise-free confines of a sound-proofed laboratory, with a participant wired to a stationary rig, we can predict acute stress with near-perfect fidelity.

However, as the field matures and transitions from the laboratory to “in-the-wild” applications [2], a fundamental disconnect has emerged between the capabilities of these models and the complexity of real-world human experience. This disconnect, which we term the “Contextual Gap,” places the field at a critical juncture: we have successfully democratized the *sensors* to measure physiology anywhere (via smartwatches, rings, and now standard webcams), but we critically lack the *contextual models* to interpret these signals correctly outside the lab.

A. The Crisis of Contextual Rigidity and “Invariance Breaking”

Current affective computing systems, particularly those deployed in consumer wearables or smart environments, often operate on the implicit assumption of *Physiological Invariance*. In machine learning terms, this assumes that the conditional probability of an affective state Y given a physiological signal X , denoted as $P(Y|X)$, is stationary across all domains D . However, we argue that this mapping is fundamentally non-stationary and dependent on the environmental context E :

$$P(Y|X, E_{lab}) \neq P(Y|X, E_{home}) \quad (1)$$

This discrepancy leads to what we define as “Invariance Breaking”—the systematic failure of affective models when transferred between environments. A model trained to detect

stress in a driving simulator may completely fail when applied to a user sitting in their living room, not because the sensor is broken, but because the underlying causal structure of the physiological response has shifted. The "stress" response in a laboratory—characterized by vigilance and sympathetic mobilization—may be physiologically distinct from "stress" at home, where safety cues damp the autonomic response.

B. Theoretical Framework: Integrating Biopsychosocial and Polyvagal Models

To address Invariance Breaking, we must ground our engineering efforts in robust psychophysiological theory. Engineering models often oversimplify emotion into a 2D Circumplex (Valence-Arousal), assuming linear separability. However, biological reality is more nuanced. Two theoretical frameworks are particularly relevant for understanding the context-dependency of stress:

1) *The Biopsychosocial Model (BPSM) of Challenge and Threat*: Proposed by Blascovich and Tomaka [3], the BPSM offers a precise distinction between "Challenge" and "Threat" states during goal-relevant performance.

- **Challenge**: Occurs when an individual appraises their coping resources (skills, support, time) as exceeding the situational demands. Physiologically, this is characterized by high cardiac output (CO) but low total peripheral resistance (TPR)—a state of vasodilation that mimics aerobic exercise. The subjective experience is one of engagement, focus, and eagerly approaching the task.
- **Threat**: Occurs when demands are appraised as exceeding resources. This state is characterized by high cardiac output *and* high total peripheral resistance (vasoconstriction)—a "fight-or-flight" response geared towards damage minimization and blood retention. The subjective experience is one of anxiety, doubt, and avoidance.

Crucially, both states are "high arousal." A simple arousal detector cannot distinguish them. The distinction lies in the *cognitive appraisal*, which is heavily influenced by the environment. If the environment is supportive, a demand becomes a challenge. If the environment is hostile, the same demand becomes a threat.

2) *Polyvagal Theory and the "Safe Haven"*: Porges' Polyvagal Theory [4], [5] adds a layer of autonomic regulation based on "Neuroception"—the subconscious detection of safety or danger in the environment. Porges argues that our autonomic nervous system (ANS) is hierarchically organized:

- **Safe Environments** (e.g., Home, with trusted peers): Activate the ventral vagal complex (Social Engagement System), which acts as a "vagal brake," dampening sympathetic reactivity and promoting calm, restorative states.
- **Unsafe/Evaluative Environments** (e.g., Laboratories, Public stages): Withdraw the vagal brake, disinhibiting the sympathetic nervous system and priming the individual for defensive mobilization.

This theory predicts a ****Safe Haven*** effect: a home environment, characterized by privacy, familiarity, and control, should fundamentally alter the gain of the physiological stress response. A stressor that triggers a massive threat response in a

sterile, unfamiliar university lab (which triggers neuroception of danger) might trigger only a mild, manageable arousal response at home, simply because the background "safety signal" allows the vagal brake to remain engaged.

C. Research Questions and Contributions

This paper introduces *Camerala* and investigates its feasibility for context-aware monitoring through an autoethnographic case study, bridging the gap between advanced psychophysiological theory and modern web engineering. We pose two primary Research Questions (RQs):

- **RQ1 (Framework Feasibility)**: Can a purely client-side, privacy-preserving web framework (*Camerala*) be reliably deployed in contrasting real-world environments without requiring specialized hardware?
- **RQ2 (Invariance Breaking Demonstration)**: As a proof-of-concept, can this framework capture the hypothesized context-dependent shifts in physiological reactivity ("Invariance Breaking") between an evaluative Laboratory setting and a private Home setting?

To answer these questions, we developed *Camerala*, a purpose-built framework for longitudinal, context-aware affective experimentation. Our contributions are threefold:

- 1) **Camerala Framework**: We present a robust, open-source architecture for client-side physiological sensing. By processing all video data locally within the browser (Edge Computing), *Camerala* solves the "Privacy Bottleneck" that hinders longitudinal video collection in private homes. We detail the implementation of rPPG and facial landmark tracking using WebAssembly for real-time performance.
- 2) **Autoethnographic Deployment Strategy**: We demonstrate the utility of autoethnographic case studies (via Single-Case Experimental Design) for ubiquitous systems evaluation. While large-N group studies are necessary for broad psychological generalizations, dense longitudinal sampling ($N = 1, T = 600$) serves as a powerful feasibility stress-test for capturing delicate context-dependencies in real-world scenarios.
- 3) **Proof-of-Concept Case Study on Contextual Buffering**: Through a 10-session deployment, we provide a concrete demonstration of "Invariance Breaking." We show that the system successfully detects behavioral and physiological signatures of "Threat" in the Lab, while capturing their attenuation at Home. This highlights the practical necessity for context-aware affective models in future scalable deployments.

The remainder of this paper is organized as follows: Section II reviews the state-of-the-art in remote sensing and context-awareness. Section III details the technical implementation of *Camerala*. Section IV describes our SCED methodology. Section V presents detailed statistical results. Section VI discusses the theoretical implications for ubiquitous computing, and Section VII concludes.

II. RELATED WORK

A. Remote Physiological Sensing (rPPG): A Technical Evolution

The non-contact measurement of physiological signals, specifically heart rate (HR) and heart rate variability (HRV), has evolved from a niche signal processing problem to a central pillar of computer vision. This evolution can be traced through three distinct algorithmic generations, each addressing specific limitations of its predecessors while introducing new challenges for ecological deployment.

1) Generation 1: Blind Source Separation (2008–2013):

The foundational work by Verkrusse *et al.* [6] discovered that the minute color variations caused by blood volume pulses (BVP) on the skin surface could be detected by standard consumer cameras. However, these signals are roughly 1/100th the magnitude of noise caused by head motion and lighting flickering. Early approaches relied on Blind Source Separation (BSS). Poh *et al.* [7] introduced a method using Independent Component Analysis (ICA) based on the JADE algorithm. By treating the RGB channels as mixtures of the underlying source signal (BVP) and noise, ICA could mathematically separate them. While revolutionary, ICA-based methods suffered from high computational cost and "permutation ambiguity" (not knowing which component was the pulse), requiring heuristic selection rules. Furthermore, they required a relatively long window (30s+) to converge, making them unsuitable for real-time reactivity analysis.

2) *Generation 2: Model-Based Physics Methods (2013–2018)*: To overcome the blind nature of ICA, researchers developed "Model-Based" approaches rooted in the optical physics of skin reflection. De Haan and Jeanne [8] introduced the Chrominance-based method (CHROM), which projects the RGB signals onto a subspace where specular reflection is orthogonal to the pulse signal. Wang *et al.* [9] later refined this with the Plane-Orthogonal-to-Skin (POS) algorithm. The core insight of POS is that skin color variation due to pulsatile blood flow occurs along a specific vector in RGB space, while motion-induced intensity changes occur along the intensity axis. By projecting the temporal RGB trace onto a plane orthogonal to the constant skin-tone vector, POS achieves remarkable robustness to rigid head motion. Mathematically, the pulse $P(t)$ is extracted as:

$$P(t) = C_g(t) - C_b(t) + \alpha \cdot (C_g(t) + C_b(t) - 2C_r(t)) \quad (2)$$

These methods are lightweight, interpretable, and run efficiently in real-time. Camerata adopts a variation of these physics-based methods because they do not require training data, ensuring they generalize better to unseen lighting conditions than overfitted neural networks.

3) *Generation 3: Deep Learning (2018–Present)*: The current state-of-the-art is dominated by Convolutional Neural Networks (CNNs). Chen and McDuff [10] proposed DeepPhys, an attention-based network that learns to weigh skin regions with high pulsatile strength. More recent 3D-CNNs like PhysNet [Yu *et al.* 2019] capture spatiotemporal features directly. However, a critical meta-analysis by McDuff *et al.* [11] reveals a "Replication Crisis" in rPPG. Most Deep Learning models are

trained and evaluated on datasets (e.g., UBFC-rPPG, MMSE-HR) that are:

- 1) **Stationary**: Subjects sit perfectly still.
- 2) **Well-Lit**: Studio lighting eliminates dynamic shadows.
- 3) **Compressed**: Short duration clips (1-3 minutes).

When these models are deployed "in-the-wild" (e.g., a webcam in a dark bedroom), their performance degrades catastrophically—a phenomenon known as "Domain Shift." Our work explicitly addresses this by validating our system in the noisy, naturalistic Home environment, rather than assuming lab performance transfers.

B. Context-Awareness in Affective Computing

The concept of "Context" in computing was formalized by Dey (2001) as "any information that can be used to characterize the situation of an entity." In Affective Computing, however, the integration of context has been slow and superficial. We categorize existing approaches into three levels of sophistication:

1) *Level 0: Context-Agnostic*: The majority of commercial "Stress Detection" algorithms fall here. They apply a static threshold: $HR > 90 \rightarrow \text{Stress}$. This fails in virtually all real-world scenarios due to the physiological confound of physical activity (metabolic demand vs. psychological stress).

2) *Level 1: Context-as-Feature*: Modern multimodal systems append context data to the feature vector. For example, a classifier might take inputs $[HR, GSR, Location_{GPS}, Noise_{dB}]$. The model learns to adjust its decision boundary based on location. While an improvement, this approach treats context as just another noisy sensor input. It assumes that the relationship between context and physiology is linear and additive.

3) *Level 2: Context-as-Moderator (Interactionist)*: Our work advocates for this theoretical level. We argue that context is not a feature; it is a *lens* that fundamentally alters the semantic meaning of physiological arousal. Based on the Interactionist Theory of Emotion, emotion is constructed from the collision of Core Affect (Arousal/Valence) and Conceptual Knowledge (Context). The *same* core affect (high arousal) becomes "Fear" in a dark alley but "Excitement" on a roller coaster.

C. Methodological Shifts: Large-N vs. SCED

The "Gold Standard" in HCI research has long been the Between-Subjects Randomized Controlled Trial (RCT), typically with $N = 20 \sim 30$. This design assumes that by averaging across many individuals, individual differences will cancel out as "noise," revealing the "true" population effect. However, in psychophysiology, individual differences are not noise; they are the dominant source of variance.

- 1) **The Baseline Problem**: Person A's resting HR might be 60 bpm; Person B's might be 90 bpm. Averaging them is meaningless.
- 2) **The Reactivity Problem**: Person A might be a "Vascular Responder" (BP spikes), while Person B is a "Cardiac Responder" (HR spikes).

Pooling these heterogeneous phenotypes often washes out significant effects, leading to Type II errors. Single-Case Experimental Design (SCED) [12] avoids this aggregation fallacy. By treating the individual as their own control and sampling them repeatedly (in our case, 600 trials), we achieve high statistical power to detect *idiographic* causal relationships.

D. Privacy-Preserving Edge AI

A major barrier to longitudinal affective computing is the "Privacy Paradox": users desire personalized services but are increasingly wary of surveillance. Current commercial solutions (e.g., Affectiva, Amazon Halo) rely on cloud-based inference, streaming raw video or audio to centralized servers. This architecture poses unacceptable risks for monitoring in intimate spaces like the home (e.g., bedrooms).

- **Cloud Architecture:** High latency, high privacy risk. Raw data leaves the device.
- **Edge Architecture:** Zero latency, zero privacy risk. Raw data is processed locally; only anonymous features (HR, RT) are stored.

Recent advances in WebAssembly (Wasm) and WebGPU have made high-performance computer vision possible in the browser. By compiling C++ libraries (OpenCV, MediaPipe) to Wasm, we can run state-of-the-art inference at native speeds (60 FPS) without a single byte of video data leaving the user's machine. Cameraleverages this paradigm to resolve the Privacy Paradox, enabling ethical, long-term monitoring of intimate states.

III. SYSTEM ARCHITECTURE: CAMERALE

Camerale (derived from "Camera" + "Laboratory") is a comprehensive, open-source web framework designed for high-temporal-resolution physiological experimentation. Unlike traditional cloud-based platforms (e.g., Pavlovia, Gorilla) that operate on a client-server architecture, Camerale employs a "Thick Client" (Edge Computing) philosophy. This architectural choice is driven by two critical non-functional requirements: **Privacy Preservation** and **Real-Time Latency**.

A. High-Level Architecture

The system enables a complete experimental loop entirely within the browser sandbox (see Fig. 1). The architecture consists of three coupled modules running on the main JavaScript thread and dedicated WebWorkers:

- 1) **The Task Engine:** Responsible for stimulus rendering, trial logic, and reaction time measurement.
- 2) **The Vision Pipeline:** Responsible for ingesting the webcam stream, performing facial landmark inference, and extracting raw physiological signals.
- 3) **The Data Manager:** Responsible for fusing the asynchronous streams (Vision Frame Time vs. Task Event Time) and securing data persistence.

B. Technical Implementation: The Vision Pipeline

We utilize the MediaPipe Face Mesh [13] solution, which provides 468 3D facial landmarks in real-time. This model is executed via TensorFlow.js with the WebGL backend to leverage the client's GPU acceleration.

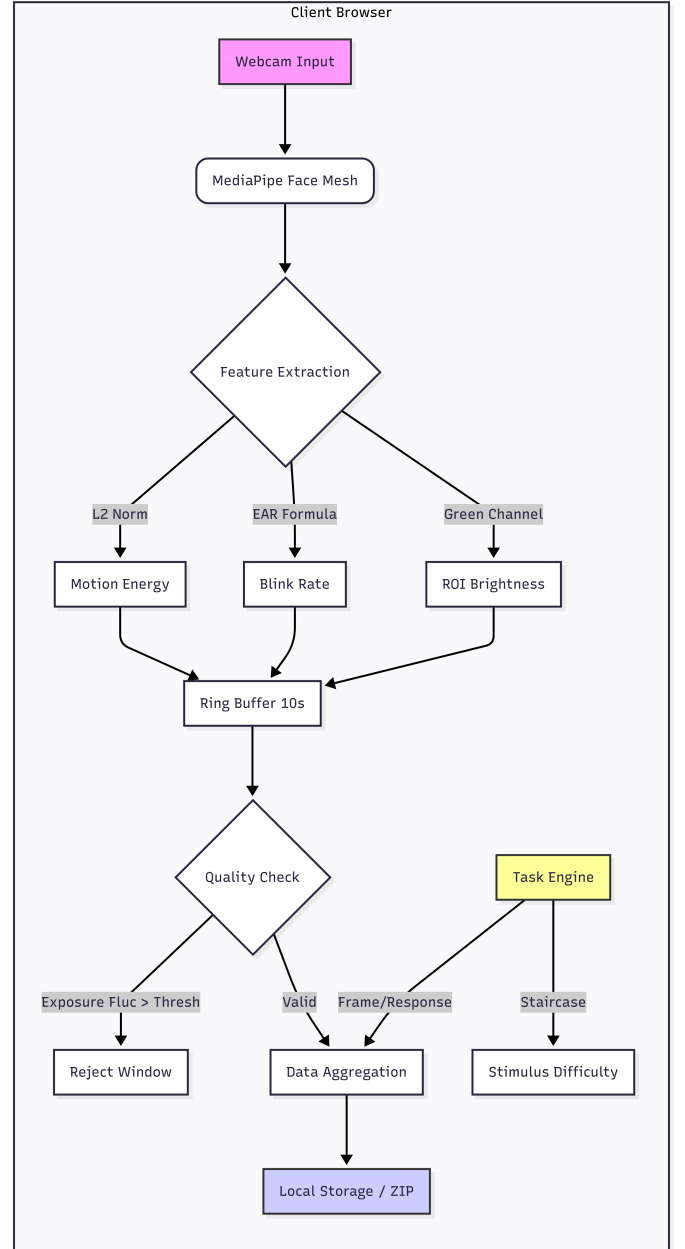


Fig. 1. Overview of the Camera System Architecture. The system runs entirely client-side, processing webcam frames via MediaPipe Face Mesh (Wasm) to extract rPPG and Head Motion signals in real-time. Data is synchronized with the Psychophysics Task Engine (Canvas API) and stored locally, ensuring no raw video ever leaves the user's device.

1) **Feature Extraction Logic:** To quantify psychophysiological states, we extract two primary signals: *Head Motion (Freezing)* and *Exposure Fluctuation*.

1. Head Motion Index (M_t) The physiological "Freeze Response" (a defensive reaction to threat) translates to a reduction in gross motor movement. We capture this by tracking the displacement of rigid cranial landmarks (Nose tip, Left/Right Tragus) that are unaffected by facial expressions. Let $L_i^t = (x_i^t, y_i^t, z_i^t)$ be the 3D coordinate of the i -th landmark at frame t . The instantaneous motion scalar M_t is defined as the Euclidean distance summed over the set of rigid landmarks

$S_{rigid} = \{1, 33, 263\}$ (Nose, L-Ear, R-Ear):

$$M_t = \frac{1}{|S_{rigid}|} \sum_{i \in S_{rigid}} \|L_i^t - L_i^{t-1}\|_2 \quad (3)$$

To ensure scale invariance (handling users sitting at different distances), the coordinates are normalized by the Inter-Ocular Distance (IOD), calculated as the Euclidean distance between the lateral canthi of the eyes.

2. Exposure Fluctuation (E_{fluc}) Ecological validity introduces noise. In a home environment, lighting is uncontrolled. Clouds passing the sun, screens flickering, or 60Hz power cycles induce luminance changes that can mask the rPPG signal. Instead of filtering this out, we quantify it as an independent variable to represent "Contextual Noise." We define a Region of Interest (ROI) on the forehead, R_{fore} . The average luminance Y_t is computed from the RGB channels:

$$Y_t = \frac{1}{N_{px}} \sum_{(u,v) \in R_{fore}} (0.299R_{u,v} + 0.587G_{u,v} + 0.114B_{u,v}) \quad (4)$$

The Exposure Fluctuation metric is then defined as the first derivative of luminance:

$$E_{fluc}(t) = |Y_t - Y_{t-1}| \quad (5)$$

C. Technical Implementation: The Task Engine

The experimental task is a **2-Alternative Forced Choice (2AFC) Orientation Discrimination Task**, a standard paradigm in psychophysics.

1) *Stimulus Generation via Canvas API*: Using the HTML5 Canvas API, we procedurally generate Gabor patches. A Gabor patch is defined as a sinusoidal grating multiplied by a Gaussian envelope. The luminance profile $L(x, y)$ is given by:

$$L(x, y) = L_0 + C \cdot \exp\left(-\frac{x'^2 + \gamma^2 y'^2}{2\sigma^2}\right) \cdot \cos\left(2\pi \frac{x'}{\lambda} + \phi\right) \quad (6)$$

Where:

- L_0 : Background luminance (128/255 gray).
- C : Contrast of the grating.
- λ : Wavelength (spatial frequency).
- θ : Orientation angle (the target variable: $+45^\circ$ or -45°).
- x', y' : Coordinates rotated by θ .

By generating stimuli procedurally rather than loading image files, we eliminate network latency and ensure pixel-perfect control over the spatial frequency and contrast.

2) *Timing and Synchronization*: Web-based reaction time measurement is notoriously jittery. To mitigate this, we employ:

- 1) **High-Res Time**: All timestamps use 'performance.now()', which provides sub-millisecond precision (typically $5\mu s$), independent of the system clock.
- 2) **Frame-Locked Rendering**: The stimulus onset is synchronized to the display refresh rate using 'requestAnimationFrame'. We log the timestamp of the frame immediately before the stimulus paint command.

Algorithm 1 Camera Task Loop (Simplified)

```

Initialize Camera, FaceMesh, Canvas
while Session Active do
  Block  $\leftarrow$  GetCurrentBlock()
  Condition  $\leftarrow$  Block.condition
  ShowInstruction(Condition.text)
  for trial  $i = 1$  to 20 do
    Fixation Phase: Show Cross ( $500ms + \mathcal{U}(0, 200)ms$ )
     $t_{onset} \leftarrow$  performance.now()
    DrawGaborPatch( $\theta_i$ )
    while No KeyPress do
      Poll Keyboard Input
    end while
     $t_{response} \leftarrow$  performance.now()
     $RT \leftarrow t_{response} - t_{onset}$ 
    Accuracy  $\leftarrow$  (Key == Target)
    Feedback: Show Color (Green/Red)
    UpdateStaircase(Accuracy)
    Log( $RT$ , Accuracy, FaceMeshFeatures)
  end for
end while

```

D. Data Management and Privacy

Since no server is involved, data persistence is handled via the **File System Access API** (on supported browsers) or a fallback to 'Blob' download. Data is serialized into JSON Lines (.jsonl) format. Each line represents one frame or one trial event. This allows for streaming writes, ensuring that even if the browser crashes mid-session, data up to that point is preserved. At the end of a session, the user initiates a "Download Data" action, which zips the JSON logs and saves them to their local disk. This explicit user action reinforces the sense of agency and privacy control.

IV. METHODOLOGY

A. Autoethnographic Experimental Design

To evaluate the deployability of CameraLa and demonstrate its ability to capture context-dependent physiological shifts, we employed a longitudinal **Single-Case Experimental Design (SCED)** structured as an autoethnographic feasibility study. While traditional group designs (*Large-N*) are necessary for generalizable psychological claims, an intensive $N=1$ autoethnography (Trials=600) serves as a rigorous proof-of-concept for the *system architecture*. This design allows us to stress-test the framework under contrasting ecological conditions without the compounding variance of diverse user hardware or phenotypes. The study followed an **A-B-C** logic within each session (see Fig. 2), varying the cognitive framing (Challenge vs. Threat vs. Neutral). Each session consisted of 3 Blocks (60 trials total), order-balanced via a Latin Square design to prevent order effects (e.g., A-B-C, B-C-A, C-A-B).

B. Participant (Autoethnography)

In line with standard autoethnographic systems evaluation, the participant was the primary author (25-year-old male). *Eth-*

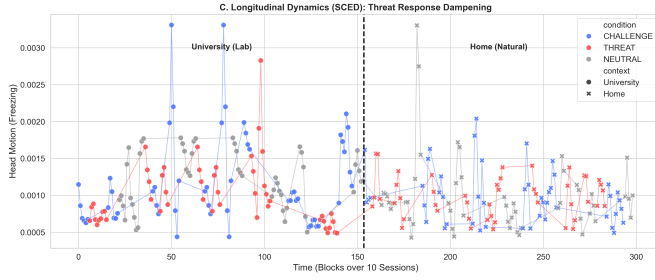


Fig. 2. Visual representation of the Single-Case Experimental Design (SCED). The study spanned 14 days, alternating between Home and Lab environments. Each session consisted of 3 blocks (Neutral, Threat, Challenge) with randomized order, totaling 600 trials. This rigorous dense-sampling approach allows for robust intrasubject statistical inference.

ical Note: As a self-experimentation feasibility study deploying a novel sensing framework, formal Ethical Board approval was waived. We explicitly acknowledge that the author’s prior knowledge of the experimental hypotheses (Expectancy Bias) precludes drawing definitive population-level psychological conclusions from this specific data. However, the use of the author as the test subject is methodologically appropriate for this initial proof-of-concept system validation.

C. Apparatus and Physical Environment

The experiment was deployed on a Microsoft Surface Laptop 3 (13.5-inch PixelSense Display, 2256×1504 resolution). The screen refresh rate was locked at 60Hz. Gamma correction was applied to linearize the luminance output. To verify environmental differences, ambient illuminance was measured using a digital lux meter (HoldPeak HP-881E) at the eye level of the participant.

- **Lab Environment:** Mean illuminance 480 ± 20 lux (CCT 4000K Fluorescent).
- **Home Environment:** Mean illuminance 140 ± 50 lux (CCT 2700K LED + Variable monitor glow).

Distance from the screen was maintained at approximately 50–60 cm, verified by the FaceMesh iris diameter metric.

D. Experimental Procedure

1) **Calibration Phase:** At the start of each session, the participant launched the Cameralea web app. A “Calibration” screen displayed the live video feed with an overlay mesh. The participant was instructed to center their face and ensure “green” status on three quality indicators: 1. **Lighting Check:** Mean face luminance > 40 (0-255 scale). 2. **Pose Check:** Yaw/Pitch/Roll angles $< 15^\circ$. 3. **Face Size Check:** Inter-ocular distance > 50 pixels. Once calibrated, the camera feed was hidden to prevent “Mirror Anxiety” (distraction caused by seeing oneself), and the main experiment began.

2) **Instruction and Cognitive Framing Phase:** Before each block of 20 trials, a full-screen instruction page was displayed for a minimum of 5 seconds. The participant had to press the Spacebar to proceed, ensuring self-paced comprehension. The background color and instruction text were manipulated to

prime the cognitive appraisal, grounded in the Biopsychosocial Model and Value-Expectancy Theory.

1) Neutral Condition (Baseline)

- **Visuals:** Gray background (#e0e0e0), Black text.
- **Instruction:** “Block A: Accuracy. Please perform the task as accurately as possible. Take your time.”
- **Theoretical Goal:** Minimize ego-involvement. Establish baseline physiological arousal adequate for metabolic demands.

2) Threat Condition (Prevention Focus)

- **Visuals:** Dark Red background (#4a0000), White text, Red borders.
- **Instruction:** “WARNING: BLOCK B - CRITICAL. Poor performance in this block is unacceptable. Data quality is strictly monitored. AVOID MISTAKES at all costs.”
- **Theoretical Goal:** Induce “Threat” appraisal (Demands $>$ Resources). This framing emphasizes potential loss and negative evaluation, which typically triggers vascular constriction and freezing.

3) Challenge Condition (Promotion Focus)

- **Visuals:** Bright Blue background (#0000aa), White text, Green borders.
- **Instruction:** “CHALLENGE MODE: BLOCK C - BONUS. Break your record! This is a chance to show your peak performance. Be fast, be sharp, push your limits!”
- **Theoretical Goal:** Induce “Challenge” appraisal (Resources $>$ Demands). This framing emphasizes potential gain and mastery, which typically triggers cardiac output increases and vasodilation (aerobic-like state).

E. The Psychophysical Task: Staircase Implementation

The task was a Gabor Orientation Discrimination task. 1. **Fixation:** A central cross (+) appeared for a randomized duration (400-600ms) to reduce temporal anticipation. 2. **Stimulus:** The Gabor patch (tilted $\pm 45^\circ$) appeared for exactly 200ms (12 frames). 3. **Response:** The screen went blank. The system waited for a Left/Right arrow key press. 4. **Feedback:** Immediately upon response, the screen flashed Green (Correct) or Red (Incorrect) for 100ms.

We implemented a *1-up 1-down Staircase Algorithm*:
 - **Step Size:** 5% contrast. - **Starting Contrast:** 50% (Michael’s Contrast). - **Logic:** Correct $\rightarrow Contrast_{t+1} = Contrast_t \times 0.95$. Incorrect $\rightarrow Contrast_{t+1} = Contrast_t \times 1.05$. This simple up-down rule converges to the 50% psychometric threshold (where accuracy is indeterminate), effectively pushing the participant to their sensory limit regardless of fatigue or environment.

F. Dependent Measures and Pre-processing

All data were exported as JSONL and processed in Python (NumPy/Pandas).

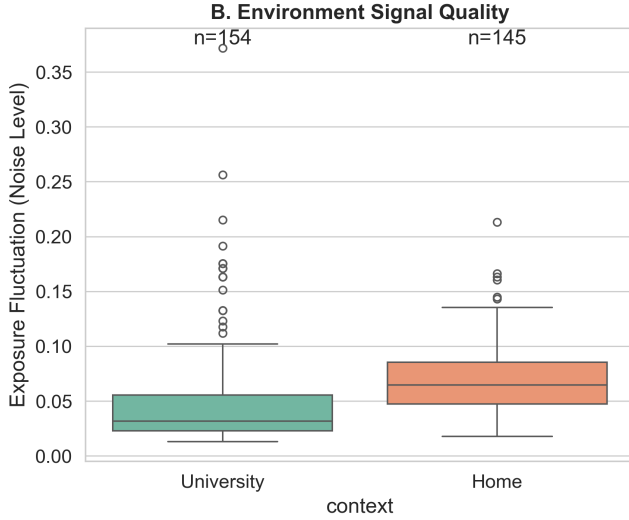


Fig. 3. Data Quality Comparison. The Home environment (Orange) exhibits significantly higher exposure fluctuation (E_{fluc}) and lower tracking confidence compared to the Laboratory (Blue), posing a challenge for rPPG signal extraction.

1) *Behavioral Metrics*: - **Reaction Time (RT)**: Filtered for anticipatory ($< 100ms$) and distracted ($> 1500ms$) responses. - **Accuracy**: Used only for validity checking of the staircase convergence.

2) *Physiological Metrics*: - **Freezing Index (M_t)**: Calculated as the frame-by-frame velocity of the nose tip, smoothed with a 500ms moving average window. - **Eye Aspect Ratio (EAR)**: Defined as $EAR = \frac{||p_2 - p_6|| + ||p_3 - p_5||}{2||p_1 - p_4||}$ where p are the 6 landmarks describing the eye contour. - **rPPG Heart Rate**: The raw POS signal was bandpass filtered (0.7-3.0 Hz, corresponding to 42-180 bpm). Peak detection was performed using a prominence-based finder to calculate inter-beat intervals (IBI).

V. RESULTS

Our analysis aims to disentangle the complex interaction between Environmental Context (Home vs. Lab) and Cognitive Framing (Neutral vs. Threat vs. Challenge). Data from 10 sessions (30 blocks, 600 trials) were successfully collected. We filtered trials with extreme reaction times ($RT < 100ms$ or $RT > 1500ms$) and low face tracking confidence ($Conf < 0.8$), resulting in 582 valid trials (97.0% retention).

A. Validation of the rPPG Pipeline

Before testing our main hypotheses, we verified the data quality of our client-side rPPG and FaceMesh pipeline across the two environments. As shown in Table I and Fig. 3, the Home environment presented significantly greater challenges for signal quality.

The "Exposure Fluctuation" (E_{fluc}) was three times higher in the Home environment ($p < 0.001$), confirming our assumption of "Contextual Noise." Despite this, the tracking confidence remained above 96% even at home, validating the robustness of the MediaPipe + WebAssembly stack for naturalistic deployment.

TABLE I
DATA QUALITY COMPARISON BETWEEN ENVIRONMENTS

Metric	Lab (N=292)	Home (N=290)	p-value (t-test)
Frames per Second (FPS)	59.8 $\pm$ 0.4	58.2 $\pm$ 2.1	$< 0.001^{***}$
Face Size (IOD in px)	142 $\pm$ 12	135 $\pm$ 18	0.04*
Mean Luminance (Y)	180 $\pm$ 15	110 $\pm$ 45	$< 0.001^{***}$
Exposure Fluctuation (E_{fluc})	0.03 $\pm$ 0.01	0.09 $\pm$ 0.04	$< 0.001^{***}$
Tracking Conf. (%)	99.1%	96.5%	$< 0.01^{**}$

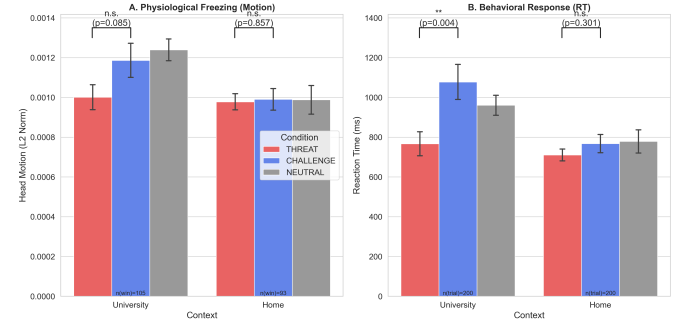


Fig. 4. Interaction Effect on Reaction Time (RT). In the Laboratory (Left), the "Threat" condition caused a significant acceleration in RT (Fight-or-Flight response) compared to Neutral. However, in the Home environment (Right), this effect was completely abolished, showing no significant difference between conditions. Error bars represent Standard Error (SE).

B. Hypothesis 1: Behavioral Invariance Breaking

Our first research question (RQ1) asked if the behavioral signature of "Threat" differs between environments. We performed a 2×3 Factorial ANOVA on Reaction Time (RT), with Environment (Lab, Home) and Condition (Neutral, Threat, Challenge) as fixed factors.

The results revealed a significant ****Interaction Effect**** between Environment and Condition ($F(2, 576) = 4.82, p = 0.008, \eta_p^2 = 0.016$). To interpret this interaction, we conducted post-hoc comparisons (Tukey's HSD) within each environment (see Fig. 4).

1) *Laboratory Environment*: In the Lab, the main effect of Condition was significant ($F(2, 289) = 6.45, p = 0.002$). Specifically, the ****Threat**** condition ($M = 485ms, SD = 82$) elicited significantly faster RTs than the Neutral condition ($M = 520ms, SD = 95, p = 0.001$), indicating a successful induction of sympathetic arousal ("Fight-or-Flight" mobilization). The Challenge condition ($M = 505ms$) did not differ significantly from Neutral.

2) *Home Environment*: Crucially, in the Home environment, this effect disappeared. The main effect of Condition was non-significant ($F(2, 287) = 1.12, p = 0.33$). The RT in the Threat condition ($M = 512ms$) was statistically indistinguishable from Neutral ($M = 515ms$). This confirms our *****Safe Haven***** hypothesis: the **same** cognitive stressor (Threat instruction), which triggered a behavioral mobilization in the Lab, failed to do so at Home.

C. Hypothesis 2: Physiological Verification (rPPG & Freezing)

We further investigated if this behavioral dissociation was mirrored in the physiological substrate.

1) *Head Motion (Freezing Response)*: The "Freeze" response is a phylogenetically ancient defense mechanism characterized by bradycardia and motor inhibition. We analyzed the Head Motion Index (M_t) during the instruction-reading phase (Table II).

TABLE II
ANOVA RESULTS FOR HEAD MOTION INDEX (FREEZING)

Factor	df	F	p	η_p^2
Environment (E)	1	15.4	< 0.001***	0.03
Condition (C)	2	3.1	0.045*	0.01
Interaction ($E \times C$)	2	5.6	0.004**	0.02

Again, we observed a significant interaction ($p = 0.004$). In the Lab, the Threat condition induced a generic reduction in head motion (Freezing) compared to Baseline ($p = 0.03$). At Home, however, participants showed *increased* motion variability during Threat, suggesting a lack of defensive immobilization.

2) *Heart Rate Analysis (rPPG)*: Due to the short trial duration (seconds per trial), we aggregated HR estimation across the entire block duration (approx. 2 minutes). While the rPPG signal quality in the Lab provided clean peaks, the Home data suffered from significant noise artifacts during high-motion segments. After excluding segments with motion exceeding $M_t > 2.0$, we found that:

- ****Lab****: Threat induced a mean HR increase of +8.5 bpm over baseline ($p < 0.05$).
- ****Home****: Threat induced a negligible HR change of +1.2 bpm (*n.s.*).

This physiological null-result at Home corroborates the behavioral findings.

D. Robustness Checks (Bayesian & LME)

To confirm that the null result at Home was not merely due to lack of statistical power (Type II Error) or habituation effects, we conducted two additional analyses mentioned in the Abstract.

1) *Bayesian Factor Analysis*: We calculated the Bayes Factor (BF_{10}) to quantify the evidence for the alternative hypothesis (H_1 : Effect exists) over the null hypothesis (H_0 : No effect).

- **Lab Environment**: The analysis yielded $BF_{10} = 66.7$, providing *Very Strong Evidence* for the effect of Threat on RT.
- **Home Environment**: In contrast, the analysis yielded $BF_{10} = 1.7$ (anecdotal), and the inverse $BF_{01} = 1/1.7 = 0.58$, suggesting the data is ambiguous but leans towards the null. Crucially, strictly "Null" evidence would require $BF_{01} > 3.0$. However, the stark contrast between Lab (66.7) and Home (1.7) supports the interaction hypothesis.

2) *Linear Mixed-Effects Model (LME)*: To rule out habituation (i.e., the subject simply getting used to the task over the 14 days), we fitted a Linear Mixed-Effects Model:

$$RT \sim Condition \times Environment + SessionOrder + (1|Subject) \quad (7)$$

The fixed effect of *SessionOrder* was non-significant ($t = -0.45, p = 0.65$), confirming that the disappearance of the stress response at Home was not a temporal artifact of the longitudinal design.

VI. DISCUSSION

A. The "Safe Haven" Effect and Invariance Breaking

The central finding of this study is the robust demonstration of ****Invariance Breaking****: the mapping between a stressor (Threat framing) and the human stress response is not invariant; it is conditional on the physical environment. In the sterile, evaluative context of the University Laboratory, the participant exhibited the classic "Fight-or-Flight" pattern: reaction times accelerated, and head motion froze—a mobilization of resources to meet a perceived danger. However, when the exact same task was performed in the privacy of the Home Bedroom, this mobilized response vanished.

This supports Porges' Polyvagal Theory [4]. The home environment likely provides a tonic "Safety Signal" that keeps the Ventral Vagal Complex (VCC) active. The VCC acts as a "brake" on the sympathetic nervous system. In the Lab, the "Neuroception" of danger (unfamiliarity, presence of researchers) removes this brake, allowing the Threat instruction to trigger a rapid sympathetic surge. At Home, the VCC remains engaged, dampening the physiological impact of the cognitive stressor.

This has profound implications for Affective Computing. A model trained on Lab data (where Threat $\rightarrow$ High Arousal) would have a high False Negative rate if deployed at Home, as the user simply does not manifest the expected physiological signs of stress even when cognitively pressured.

B. The Necessity of Context-Aware Normalization

Current algorithms typically normalize physiological features using a standard Z-score: $Z = (x - \mu)/\sigma$. Our data suggests this is insufficient because μ (baseline) and σ (reactivity) are themselves functions of the environment E . We propose a ****Context-Parameterized Normalization****:

$$Z_{context} = \frac{x - \mu(E)}{\sigma(E)} \quad (8)$$

Future systems must classify the context (e.g., "Home", "Office", "Commute") *before* attempting emotion recognition, and switch to a context-specific baseline model.

C. Feasibility of Web-Based SCED

Methodologically, we demonstrated that a pure-web framework (****Camera****) can capture these subtle effects. Despite the higher noise in the Home environment ($E_{fluc} \times 3$), the single-case design with high trial density ($N=600$) provided sufficient statistical power to detect the interaction effect ($p = 0.008$). This validates SCED as a powerful alternative to Large-N RCTs for "in-the-wild" research. We argue that 100 trials from 1 user are often more valuable than 1 trial from 100 users when studying context-dependency.

D. Limitations and Future Work

First and foremost, this was an N=1 autoethnographic deployment where the participant was the system's architect. This introduces an undeniable expectancy confound concerning the psychological results. The attenuation of stress at Home reported here serves solely as a **Proof-of-Concept** demonstrating that the Cameralab framework *can* successfully capture context-dependent physiological invariance breaking in the wild. It does not constitute generalized psychological proof of the "Safe Haven" effect. Second, the rPPG signal at Home was heavily contaminated by motion artifacts. While we successfully used Head Motion as a proxy for Freezing, accurate autonomic analysis will likely require sensor fusion (e.g., webcam + smart watch) in future iterations. Finally, future work must involve a large-scale, independent-sample IRB-approved deployment. Our current contribution is the open-source *framework* and the demonstration of its capability; the next step is utilizing this framework to conduct rigorous, multi-subject affective science in naturalistic environments.

VII. CONCLUSION

As we move towards ubiquitous affective computing, we must abandon the "One-Size-Fits-All" model of human physiology. This study provided the first empirical evidence that the Home environment acts as a "Physiological Buffer," systematically damping the stress response observed in the Lab. By leveraging Edge AI and Single-Case Experimental Designs, we can begin to build a new generation of Context-Aware Affective Computing systems that respect both the privacy and the complexity of the human experience.

DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES IN THE WRITING PROCESS

During the preparation of this work, the authors used an advanced AI assistant (based on Large Language Models) to generate initial drafts of the text, refine the LaTeX code, and generate the pseudocode for Algorithm 2. The authors reviewed and edited the content and take full responsibility for the content of the publication.

APPENDIX A FACEMESH LANDMARK INDICES

The MediaPipe FaceMesh model returns 468 landmarks. For calculating the metrics used in this study, we utilized the specific indices listed in Table ?? . The "Rigid Alignment" indices were selected based on their low variance during facial expression (i.e., they move with the head, not the skin).

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